

Classification of postures of people with hand movement working in office environment by LiDAR sensor network

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ABSTRACT An edge computing system is proposed for classifying postures with hand movement of office workers, a remaining issue in prior work, using a light detection-and-ranging (LiDAR) sensor network. The LiDAR sensor network acquires human poses with hand movement; preprocessing includes trimming, feature estimation, clustering, and normalization before deep-learning classification. Experiments capture posture data from multiple subjects using multiple LiDAR sensors and vary data type, LiDAR placement, features, and clustering methods. Classification accuracy and confusion matrices are reported to compare preprocessing configurations. The results show effective classification under various preprocessing configurations.

INDEX TERMS posture classification, light detection and ranging, point cloud, deep learning.

I. INTRODUCTION

THE office-efficiency monitoring has been attracting people [1]–[6]. A recent report indicates that the main functions of office-efficiency monitoring include productivity monitoring, project tracking, and employee monitoring [1]. One popular system for analyzing the activities of workers in an office utilizes sensory data collected from the office environment as input data and implements fuzzy finite state machines to model the behavior of the office workers [7]. In another system, environmental variables and the personal physical and mental parameters of employees in different office environments are measured and compared to determine their correlation with productivity [8]. Biometric data and workplace environmental data are captured through micro-surveys on mobile and web applications [8].

The office-efficiency monitoring requires recording activities of workers in an office that change over time. It is insufficient to monitor only gross postures or presence. Hand-movement classification is necessary for recording the office activities. Desk work is central to the office activities. The desk work is performed through hand-involved movements. The hand-involved movements cannot be distinguished by the gross postures or presence alone. Fine-grained classification

of hand movements distinguishes hand-involved work states. The hand-movement classification can distinguish elements of the office activities as follows. The distinction supports identification of task switching. The distinction supports estimation of actual work duration.

Prior works focused on identifying human body orientation and general human postures. Katayama et al. reported that point clouds can be utilized to classify human posture and presented a system for posture recognition and tracking [9]. Their system acquired point cloud data with a small light detection-and-ranging (LiDAR) unit and classified posture using machine learning on data from a sitting person [9]. Human orientation estimation was evaluated utilizing two-dimensional convolutional neural network (2D-CNN), not hand-involved desk postures. The experiments included only one participant. Glandon et al. reported that point clouds can be utilized for human identification and presented a system for posture estimation using data captured from walking people [10]. Their system extracted a silhouette, estimated body joints, and classified posture from joint features using 3D LiDAR videos of walking subjects [10]. Glandon et al. targeted locomotion posture of walking subjects rather than seated hand movements at a desk. The weakness of the prior

works is that they cannot perform the hand-movement classification necessary to distinguish elements of the office activities. Hand-movement classification requires a classification model that extracts point-cloud details and handles multiple people.

An edge computing system is proposed that classifies the postures of people with hand movement in an office by using a LiDAR sensor network. The system captures, stores, and classifies human postures with hand movement from the LiDAR sensor network. Experiments use multiple LiDAR sensors to capture posture patterns from multiple subjects; preprocessing and deep-learning classification vary data type, LiDAR placement, features, and clustering method. The evaluation metric is the ratio of samples correctly classified. Confusion matrices for selected models are presented to show correct-label counts and biases in inferred labels.

Section II summarizes related methods. Section III presents the system model and methodology. Section IV describes the experiment setup. Section V reports the results. Section VI summarizes the findings.

II. RELATED WORKS

A. MODEL ARCHITECTURE

Classification estimates which category a given data sample belongs to by associating it with the highest estimated probability. Deep learning models for point cloud classification have attracted interest [11].

Early approaches relied on voxelization [12]–[14]. Qi et al. presented PointNet [11]. PointNet applies shared multilayer perceptrons (MLPs) to each point and aggregates per-point features using max pooling with permutation invariance.

Subsequent CNN-based models include PointCNN with permutation-invariant X-Conv [15], ShellNet with concentric-shell ShellConv [16], and A-CNN with annular convolutions on sampled neighborhoods [17].

Other approaches encode spatial structure with self-organizing maps in SO-Net [18], [19] or kd-trees in Kd-network [20].

The existing research shown above can be broadly classified into the following two categories:

- 1) Architectures for direct point cloud classification (e.g., PointNet, PointCNN).
- 2) Spatial modeling techniques using local relationships (e.g., ShellNet, SO-Net, A-CNN, Kd-network).

The present study targets fine-grained hand-movement classification in a multi-person office environment. The present study differs from the macro-posture or static object classification in the categories above. PointNet++ is adopted in the proposed system because hierarchical grouping extracts local geometry from sparse desk-region point clouds without voxelization [21].

B. IMPROVEMENT FOR OBJECT DETECTION

Point features encode relationships between points and local shape characteristics and are used as inputs to machine learning along with point coordinates [22]–[24].

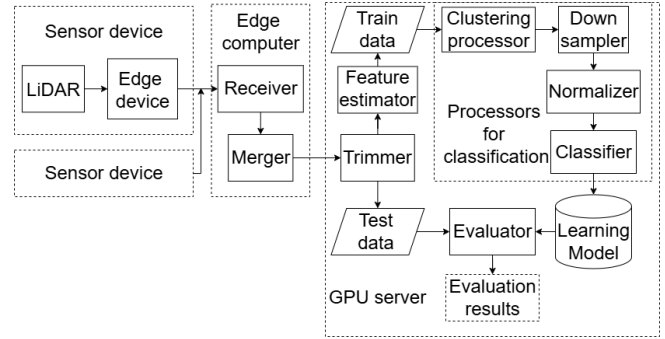


FIGURE 1. System model.

Eigenvalue-based descriptors derive features such as eigenvalue entropy and omnivariance from neighborhood covariance matrices [22]–[24]. Eigenvalue entropy indicates local orderliness of a point cloud region. Omnivariance separates linear structure points from surrounding objects.

Convex-hull descriptors assign features based on each point's position relative to the hull [25].

Higher-dimensional descriptors include Signature of Histograms of Orientations (SHOT) [26] and Position and Orientation Distribution (POD) [27] for object recognition and 3D shape retrieval.

The existing research shown above focuses on geometric feature descriptors for shape recognition. The present study applies point feature preprocessing to support fine-grained hand movement classification in a multi-person office environment. The existing research above targets static object shape recognition rather than fine-grained dynamic hand movements. The proposed system evaluates normals, dimensionality features, proportion of variance, and fast point feature histograms (FPFH) because each descriptor encodes a distinct aspect of local hand-region geometry relevant to desk postures.

III. PROPOSED SYSTEM

A. SYSTEM MODEL

Fig. 1 shows the system model of the proposed system, comprising sensor devices, an edge computer, and a GPU server. Each sensor device includes a LiDAR unit and an edge device; the edge computer receives and merges streams from multiple devices; the GPU server performs trimming, feature estimation, clustering, downsampling, normalization, and classification. Sensor-side storage and edge-side merging reduce data transfer before GPU-side preprocessing and classification.

The LiDAR unit acquires 3D data stored by the edge device; the edge computer merges received streams; the GPU server trims each frame to the person region, extracts point features, clusters and downsamples the point cloud, normalizes coordinates, and classifies postures.

B. METHODOLOGY

3D data is acquired using the system presented by Akiyama et al. and Li et al. [28], [29]. Multiple LiDAR calibrations merge streams by rotating and translating 3D data according to predetermined angles and distances.

The point cloud of each frame of the aggregated 3D data is trimmed. Vertices of each person's region are determined manually under a fixed-seating office layout.

Four feature types were selected to capture complementary geometric properties of hand-region point clouds. Normals encode local surface orientation of hands and forearms. Dimensionality features distinguish linear, planar, and volumetric local structures. Proportion of variance summarizes eigenvalue distribution within each neighborhood. FPFH encodes pairwise angular relationships among neighboring points. Four types of features are extracted from the trimmed point clouds: normals, dimensionality features, proportion of variance, and FPFH. For normals, dimensionality features, and proportion of variance, the k-nearest neighbor algorithm (KNN) uses search radius 100 and maximum 30 neighbors to build covariance matrices at each point. *Normals*. Eigenvectors of the smallest eigenvalue in each covariance matrix serve as normals (Fig. 2(a)). *Dimensionality features* [30], [31]. Eigenvalues λ_i , $i = [1, 3]$ yield $a_{1D} = \frac{\sqrt{\lambda_1 - \lambda_2}}{\sqrt{\lambda_1}}$, $a_{2D} = \frac{\sqrt{\lambda_2 - \lambda_3}}{\sqrt{\lambda_1}}$, and $a_{3D} = \frac{\sqrt{\lambda_3}}{\sqrt{\lambda_1}}$ [30], [31] (Fig. 2(b)). *Proportion of variance* [32]. Proportion of variance is $v_i = \frac{\lambda_i}{\lambda_1 + \lambda_2 + \lambda_3}$ ($i = [1, 3]$) (Fig. 2(c)). *FPFH* [33]. Two adjacent points per center point define a Darboux frame; FPFH is computed from frame-based angular relations and concatenated; principal component analysis (PCA) with randomized singular value decomposition reduces the 33-dimensional FPFH to three dimensions (Fig. 2(d)).

Next, clustering is performed for the point clouds using the point features. DBSCAN ($\epsilon = 100$, minimum 10 points on 3D coordinates) and k-means ($k = 3$) each remove outliers and extraneous clusters [34], [35]. DBSCAN removes isolated noise points from background objects and sensor reflections. k-means partitions the point cloud into fixed spatial groups; extraneous partitions are discarded. Aggressive cluster removal can delete sparse finger points and degrade classification of fine hand postures.

Downsampling is performed for the clustered point clouds.

Downsampled training point clouds are normalized so that the distance from the center of gravity to the farthest point equals 1.0.

PointNet++ classifies normalized point clouds by repeatedly selecting points, grouping neighborhoods, and processing each group with PointNet and a fully connected layer [21]. Two-level grouping in PointNet++ captures hand-scale local structures that distinguish mouse operation, pad operation, typing, and sitting still.

IV. EXPERIMENT SETUP

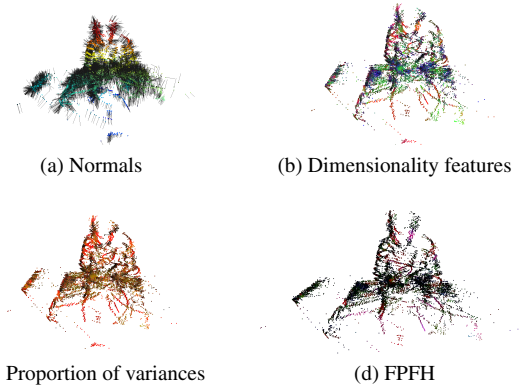


FIGURE 2. Point clouds with each of four types of feature.

A. EXPERIMENTAL SYSTEM

An experimental system was built from the model in Fig. 1. The experimental system consisted of two sets of LiDAR units (Mid-70 units), two edge devices (Jetson nano units), an edge computer (Jetson Xavier NX and Jetson Orin nano), and a regular keyboard [36]–[39]. The experiment was conducted at 14Q32, 14F, Shibaura Institute of Technology, Toyosu, Koto-ku, Tokyo. Two Livox Mid-70 LiDAR units were deployed at a sensor frame rate of 10 fps. Two NVIDIA Jetson nano edge devices were used. An NVIDIA Jetson Xavier NX edge computer was used for training data and test data A, B, and C acquisition. An NVIDIA Jetson Orin nano edge computer was used for test data A to AB acquisition.

Training data and test data A, B, and C were acquired on 9 December 2022. Test data A to AB were acquired on 12 December 2024. 3 contributors provided training data. 28 contributors provided test data.

B. LIDAR DEPLOYMENT

The two LiDAR sensors were placed on a table facing each other. They were positioned 1.0 m away from the subject in a lateral direction and tilted down by 10° to make the PC easier to detect. The PC was placed on the table in front of the subject. Fig. 3 shows the experiment site.

C. DATA ACQUISITION

1) Training data acquisition

LiDAR was used to acquire training data on the subject's posture while using the PC to perform four tasks.¹

Table 1 summarizes data acquisition conditions for training data; each capture was repeated 20 times for mouse operation, pad operation, and typing, and 10 times for sitting still.

Fig. 4 shows point clouds for each posture class.

The three subjects in this study are denoted as subjects A, B, and C. Training data with subjects A, B, and C is called training data A, B, and C.

¹Our experiment was approved by the Ethics Committee of Shibaura Institute of Technology.



FIGURE 3. Experiment site.

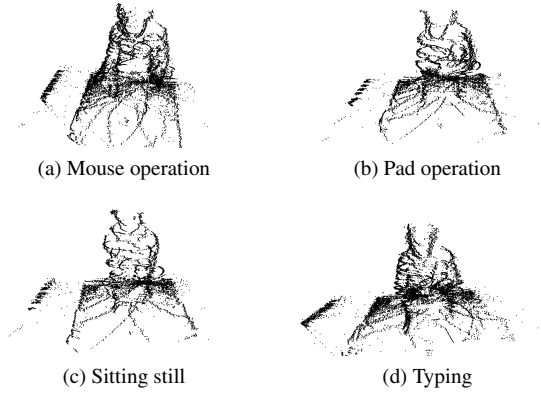


FIGURE 4. Visualization of data showing each label.

2) Test data acquisition

Test data were acquired for the four posture classes under the conditions summarized in Table 1; each capture was performed once per subject.

The 28 subjects in this study are denoted as subjects A to AB. Test data with subjects A to AB is called test data A to AB.

D. PROCESSING OF PROPOSED SYSTEM

Processing followed Section III-B: two vertices defined each person region for trimming; Open3D extracted normals, dimensionality features, proportion of variance, and FPFH [40]. DBSCAN and k-means removed outliers as in Section III-B; point counts were adjusted to 4,096 after clustering and downsampling [41]. Normalization was performed using

TABLE 1. Data acquisition conditions for each posture class.

Posture class	Task	Duration
Mouse operation	Write alphabet A–Z using mouse	30 s
Pad operation	Write alphabet A–Z using trackpad	30 s
Sitting still	Sit still without moving or touching PC	10 s
Typing	Type alphabet A–Z on keyboard	30 s

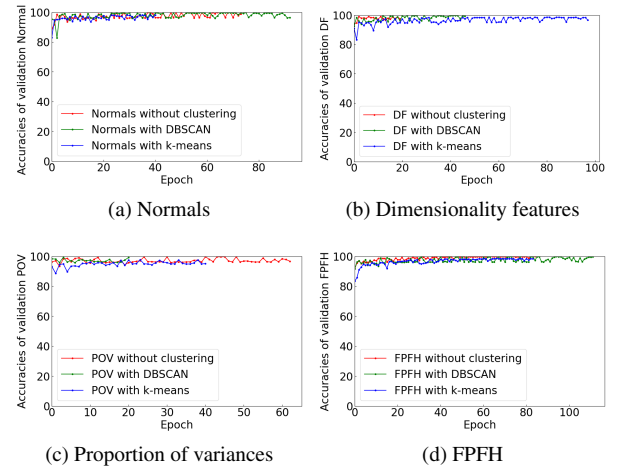


FIGURE 5. Classification accuracies of validation.

different values for each point cloud dataset used for the training. PointCutMix augmented normalized training data by replacing the top $\lambda\%$ of nearest-neighbor-matched points between three point clouds A, B, and C, with λ drawn from a beta distribution [42]. PointCutMix increases shape diversity among training samples and improves robustness to subject-dependent point-cloud density.

Data were split 4:1 for training and validation; multi-scale grouping (MSG) formed PointNet++ groups [41], [43].

E. EVALUATION METRIC

The classification models trained using the augmented data from three subjects were evaluated using the test data from twenty-eight subjects.

Accuracy was utilized to evaluate the classification models, defined as

$$\frac{TM + TP + TS + TT}{TM + TP + TS + TT + FM + FP + FS + FT}, \quad (1)$$

where TM is the number of correctly classified point clouds that actually indicate the mouse operation, TP , TS , and TT are the number of correctly classified point clouds that actually indicate the pad operation, the sitting still without moving, and the typing, respectively, and FM , FP , FS , and FT are the number of point clouds that were not correctly classified.

V. EVALUATION

A. VALIDATION RESULTS

Fig. 5 shows validation accuracy per epoch for each feature type; each subfigure plots three lines for no clustering, DBSCAN, and k-means.

For normals (Fig. 5(a)), validation accuracy remained high under all clustering settings. Normal features assign one direction per point and are less sensitive to point distribution changes caused by clustering. For dimensionality features (Fig. 5(b)), validation accuracy remained high; k-means showed a slight decrease relative to no clustering and DBSCAN. k-means can remove points that carry orientation

information needed for dimensionality features. For proportion of variance (Fig. 5(c)), validation accuracy remained high with a similar k-means decrease. For FPFH (Fig. 5(d)), validation accuracy remained high; DBSCAN showed larger epoch-to-epoch variation than the other clustering settings. DBSCAN varies cluster counts across frames and alters the local shape encoded by FPFH.

Training terminated if validation accuracy did not improve within 15 epochs after the best epoch.

B. ACCURACY

Table 2 shows the accuracy for all twelve combinations of features and clustering methods used. Subjects A to AB served as the test dataset for four posture classes; accuracy exceeded the 25% baseline in all cases. Classification accuracy varied across test subjects because point-cloud density and body size differed from the three training subjects.

Regarding the clustering methods, no clustering or DBSCAN tended to achieve higher accuracy than K-means. k-means tended to lower accuracy by removing hand-region points required to distinguish typing from pad operation.

In terms of features, normals tended to yield higher accuracy when no clustering or DBSCAN was employed. Normals yielded stable test accuracy when no clustering or DBSCAN was applied. Dimensionality features achieved higher accuracy than other features for several subjects outside the training set when local shape resembled training examples.

VI. CONCLUSION

An edge computing system was proposed that classifies the postures of people with hand movement working in offices by using a LiDAR sensor network. Fine hand movements are classified by selecting appropriate point-cloud features. The classification model focuses on features that represent point-cloud shape and local geometry.

Experiments acquired real LiDAR data from 28 test subjects and three training subjects; preprocessing and deep-learning classification varied data type, LiDAR placement, features, and clustering method.

Validation accuracy reached nearly 100% for all feature and clustering combinations, with minor differences among clustering methods. High validation accuracy indicates that PointNet++ learned discriminative patterns from the selected point features under each preprocessing configuration.

Test accuracy exceeded the 25% baseline for four posture classes across feature and clustering combinations; normals without clustering or DBSCAN yielded stable accuracy, and no clustering or DBSCAN outperformed k-means. Test accuracy above the 25% baseline confirms generalization beyond the three training subjects for four desk postures. Misclassification between typing and pad operation occurred when both hands rested near the keyboard and trackpad with similar point-cloud layouts. Clustering methods that discarded sparse hand points increased label bias toward incorrect posture classes.

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TABLE 2. Classification accuracy for each subject and feature and clustering combination.

Subject	Normals			Dim. features			Prop. variance			FPFH		
	None	DBSCAN	k-means	None	DBSCAN	k-means	None	DBSCAN	k-means	None	DBSCAN	k-means
A	0.65	0.71	0.60	0.43	0.86	0.79	0.70	0.82	0.70	0.73	0.85	0.75
B	0.98	0.99	0.88	0.94	0.97	0.92	0.98	0.91	0.82	0.99	0.96	0.89
C	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70
D	0.61	0.72	0.41	0.59	0.67	0.63	0.47	0.44	0.63	0.50	0.60	0.55
E	0.63	0.69	0.46	0.73	0.60	0.59	0.60	0.58	0.56	0.66	0.53	0.57
F	0.72	0.75	0.54	0.83	0.58	0.56	0.61	0.55	0.62	0.76	0.58	0.50
G	0.68	0.75	0.40	0.72	0.53	0.54	0.56	0.55	0.54	0.75	0.55	0.40
H	0.71	0.74	0.68	0.75	0.68	0.66	0.60	0.62	0.61	0.72	0.63	0.69
I	0.76	0.90	0.63	0.92	0.66	0.65	0.61	0.61	0.60	0.75	0.64	0.62
J	0.72	0.89	0.88	0.68	0.90	0.75	0.73	0.80	0.84	0.95	0.81	0.92
K	0.74	0.87	0.87	0.97	0.86	0.77	0.65	0.82	0.79	0.83	0.80	0.80
L	0.73	0.83	0.90	1.00	0.86	0.81	0.63	0.83	0.80	0.80	0.81	0.85
M	0.59	0.59	0.45	0.56	0.60	0.44	0.56	0.46	0.40	0.49	0.60	0.45
N	0.67	0.60	0.62	0.63	0.76	0.75	0.84	0.60	0.68	0.75	0.84	0.68
O	0.85	0.71	0.71	0.70	0.76	0.73	0.73	0.61	0.68	0.80	0.73	0.72
P	0.79	0.72	0.70	0.70	0.82	0.77	0.68	0.56	0.75	0.81	0.80	0.76
Q	0.70	0.67	0.68	0.70	0.70	0.64	0.66	0.60	0.64	0.71	0.70	0.68
R	0.90	0.88	0.92	0.72	0.99	0.84	0.84	0.89	0.78	0.80	1.00	0.87
S	0.68	0.72	0.78	0.92	0.80	0.73	0.60	0.79	0.65	0.59	0.71	0.83
T	0.93	0.87	0.70	0.70	0.79	0.67	0.79	0.70	0.71	0.85	0.91	0.74
U	0.69	0.81	0.85	0.71	0.91	0.76	0.80	0.89	0.82	0.87	0.80	0.86
V	0.62	0.62	0.82	0.74	0.64	0.66	0.61	0.57	0.70	0.54	0.57	0.66
W	0.62	0.63	0.65	0.32	0.58	0.56	0.56	0.57	0.50	0.41	0.62	0.41
X	0.61	0.65	0.47	0.36	0.64	0.48	0.53	0.60	0.47	0.41	0.57	0.43
Y	0.60	0.59	0.64	0.61	0.67	0.62	0.60	0.63	0.60	0.60	0.65	0.61
Z	0.62	0.59	0.68	0.77	0.68	0.58	0.60	0.64	0.55	0.66	0.67	0.66
AA	0.64	0.87	0.94	0.90	0.79	0.88	0.62	0.76	0.71	0.58	0.54	0.63
AB	0.70	0.90	0.96	0.82	0.94	0.87	0.68	0.94	0.83	0.87	0.72	0.86

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